FINAL CHANGE SUMMARY & PROJECT AUDIT

Group: G1 Mini-Project (ISA Term Assessment)

Project: Me

1. What Was Already Working in the Initial ZIP

- High-level layout of the Spotify-themed dark dashboard with green accents.

- Basic c

2. What Was Changed and Improved

Algorithm & Architectural Alignment

1. Academic Merge Sort Alignment (PDF Reference Notes):

- Refact
indexing

3. Bugs Fixed

1. Index Boundary Mismatch: Fixed previous half-open [0, n) range indexing to match the academic lecture notes' inclusive [low, high] bounds.
2. Missin values

4. Tests Actually Performed

All 15 tests were implemented and verified with node test_suite.js:

Test #	Description	Outcome
1	PDF Numerical Example (`310, 285, 179, ...`)	**PASSED**
2	Already-Sorted Input Preserv	**PASSED**
3	Reverse-Sorted Input	**PASSED**
4	Duplicate Values & Stable So	**PASSED**
5	Empty Input Handling	**PASSED**
6	Single Element Base Case	**PASSED**
7	Negative and Decimal Numeric Sorting	**PASSED**
8	CSV Multi-column, Quoted C	**PASSED**
9	Log Timestamps Chronological Sorting	**PASSED**
10	Descending Order Sorting	**PASSED**
11	Metrics Accuracy (Comparisons, Merges, Calls,	**PASSED**
12	Recursion Tree Nodes & Dep	**PASSED**
13	File Export Integrity (CSV & JSON Export)	**PASSED**
14	Malformed / Missing Keys Ha	**PASSED**
15	Large Dataset Scaling (1,000 Elements)	**PASSED**

5. How to Run the Final Application

Method 1: Local Browser (Recommended)

1. Double-click `index.html` in file explorer.
2. The ap

Method 2: Local Static Server (Optional)

```
npm start
# or
npx serve .
```

Method 3: Cloud Deployment

- Vercel: Import repository `PixelNinja420/Merge-Sort` into Vercel and click Deploy.
- Render

6. Generated Project Files Summary

File Name	Purpose / Description
`index.html`	Complete, zero-dependency application (HTML,
`sample_server.log`	Production server logs with ISO timestamps fo
`sample_movies.csv`	Multi-column CSV dataset with headers and mov
`sample_products.csv`	RFC-4180 test CSV with quoted commas, double
`pdf_example.txt`	The 10 numbers from the academic reference le
`sample_large_logs.log`	500-line randomized server log file for demon
`test_suite.js`	Automated 15-point test runner verifying algo
`vercel.json`	Configuration for zero-config static deployme
`render.yaml`	Configuration for static site deployment on R
`package.json`	Standard metadata and scripts for `npm start`
`README.md`	Comprehensive GitHub and developer documentat
`README.txt`	Plain-text instructions for offline viewing.
`COLLEGE_MINI_PROJECT_REPORT.md`	Formal academic mini-project submission repor
`DEMO_AND_VIVA_GUIDE.md`	2-3 minute demonstration script and 15 top vi
`FINAL_CHANGE_SUMMARY.md`	Complete changelog, bug fixes, and verificati
`mergeflow_g1_project.zip`	Self-contained offline distribution ZIP archi